

DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for High-Throughput De Novo Peptide Sequencing

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Abstract

Tandem mass spectrometry (MS/MS) enables high-throughput identification of proteins, yet identifying novel peptides without reference databases—*de novo* peptide sequencing—remains computationally challenging. Existing state-of-the-art approaches rely on autoregressive transformers that predict amino acid sequences token by token in $\mathcal{O}(L)$ sequential steps. While accurate, autoregressive decoding suffers from error propagation, directional fragmentation asymmetry between complementary *b*- and *y*-ions, and severe latency bottlenecks. Continuous normalizing flows alleviate sequential decoding but suffer from severe discretization error upon rounding continuous latent states to discrete amino acid masses.

In this work, we introduce **DFlowNovo**, a non-autoregressive generative framework that models *de novo* peptide sequencing as continuous-time Discrete Flow Matching (DFM) over the probability simplex of amino acids, coupled with exact Dynamic Programming Knapsack reachability filtering (**KnapsackDP**). DFlowNovo generates complete peptide sequences in a constant budget of $K = 20$ reverse flow steps, achieving an inference throughput of 227.1–255.8 spectra per second on an NVIDIA GPU (a $4.3\times$ to $4.9\times$ speedup over autoregressive beam search). On the standardized 104k Nine-Species test benchmark, DFlowNovo achieves 68.28% strict exact match and 83.01% amino acid F1 score, outperforming InstaNovo v1.2 (65.48%) and accepting 86,412 peptide-spectrum matches at 80% precision. Furthermore, we dissect the physical foundations of mass spectrometry, demonstrating that remaining prediction discrepancies on human data are predominantly driven by isomeric Leucine/Isoleucine equivalence (113.084 Da) which requires radical-driven *w*-ion fragmentation to break.

Résumé

La spectrométrie de masse en tandem (MS/MS) permet l'identification des protéines à haut débit, mais le séquençage *de novo*—l'identification des peptides sans base de données de référence—reste un défi informatique majeur. Les modèles autorégressifs actuels prédisent les acides aminés de manière séquentielle en $\mathcal{O}(L)$ étapes, ce qui induit une propagation des erreurs, une asymétrie entre ions complémentaires *b* et *y*, et une forte latence. Les modèles à flux continus évitent ce coût séquentiel mais souffrent d'une perte de discrétisation critique lors de la projection sur les masses discrètes des résidus.

Ce travail présente **DFlowNovo**, un cadre génératif non-autorégressif formulant le séquençage *de novo* comme un appariement de flux discret (DFM) en temps continu sur le simplexe de probabilité, guidé par une programmation dynamique exacte du sac à dos (**KnapsackDP**). DFlowNovo décode les séquences peptidiques entières en un nombre fixe de 20 pas d'intégration, atteignant une vitesse de 227 à 256 spectres par seconde (4,3 à 4,9 fois plus rapide que les modèles autorégressifs). Sur le banc d'essai biologique des Neuf Espèces (104 163 spectres), DFlowNovo atteint 68,28% d'exactitude stricte et 83,01% de score F1, surpassant InstaNovo v1.2 tout en identifiant 86 412 spectres à 80% de précision. Enfin, une analyse biophysique démontre que les ambiguïtés résiduelles découlent de l'isomérisation fondamentale Leucine/Isoleucine, indissociable par fragmentation collisionnelle standard.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.

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Joël Gédéon, September 2026

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1. Introduction to De Novo Peptide Sequencing and Mass Spectrometry

Proteins are the primary functional macromolecular effectors of biological processes, executing enzymatic catalysis, structural scaffolding, signal transduction, and immunological defense within all living organisms. Deciphering the exact amino acid sequence of proteins and peptides is therefore foundational to molecular biology, pharmacology, clinical oncology, and biotechnology. Tandem mass spectrometry (MS/MS) coupled with liquid chromatography (LC-MS/MS) has emerged as the premier high-throughput analytical technology for resolving complex cellular proteomes. However, translating high-dimensional, noisy fragmentation spectra into accurate primary sequences remains one of the most demanding inverse problems in computational biology.

1.1 The Proteomics Frontier and Tandem Mass Spectrometry

In a standard bottom-up proteomics workflow, extracted proteins are enzymatically digested into shorter peptide fragments (typically 7 to 35 amino acids in length) using endopeptidases such as trypsin (Olsen et al., 2004). These peptides are separated by reversed-phase high-performance liquid chromatography and introduced into a mass spectrometer via electrospray ionization (ESI).

During the initial survey stage (MS1), the instrument measures the mass-to-charge ratios (m/z) and isotopic abundance envelopes of intact, multiply protonated peptide precursor ions. Subsequently, selected precursor ions within narrow isolation windows are accelerated into a collision cell for tandem mass spectrometry (MS2). In Higher-Energy Collisional Dissociation (HCD) (Olsen et al., 2007), precursor ions collide with neutral inert gas molecules (such as nitrogen or argon), converting kinetic energy into vibrational excitation. This vibrational excitation induces unimolecular thermal decomposition across the weakest covalent bonds of the peptide backbone, predominantly the amide peptide bonds (Paizs and Suhai, 2005). The resulting fragment ions are injected into an ultra-high-resolution mass analyzer, such as an Orbitrap, which records their exact m/z and intensity values, generating an experimental MS2 fragmentation spectrum.

1.2 Database Search Engines versus De Novo Sequencing

For over two decades, the dominant paradigm for analyzing tandem mass spectra has been database-dependent search algorithms, pioneered by SEQUEST (Eng et al., 1994) and subsequently refined in tools such as Mascot, MaxQuant, and MSFragger. Database search engines operate by matching experimental spectra against *in silico* theoretical spectra generated from reference proteome sequence databases (such as UniProt). While highly sensitive and computationally tractable for well-characterized model organisms, database searching possesses severe intrinsic limitations (Nesvizhskii, 2014):

- (i) **Unannotated Genomes and Non-Model Species:** For the vast majority of biological species on Earth, high-quality reference genomes or proteome annotations do not exist. Database searching fails entirely when the target organism is unsequenced.
- (ii) **Cancer Neoantigens and Somatic Mutations:** Malignant cells generate novel peptide sequences resulting from point mutations, gene fusions, alternative splicing, and frame-shift events that are absent from canonical reference databases. Identifying these tumor-specific antigens is vital for personalized immunotherapy and cancer vaccine development.

- (iii) **Immunopeptidomics and Antibody Repertoires:** B-cell and T-cell receptor repertoires undergo somatic hypermutation and $V(D)J$ genetic recombination, generating hypervariable complementarity-determining regions (CDRs). Assembling monoclonal antibodies or polyclonal serum repertoires directly from biological fluids requires sequence identification without genomic references.
- (iv) **Post-Translational Modifications (PTMs):** Variable chemical modifications (phosphorylation, acetylation, ubiquitination, methylation) expand the combinatorial search space exponentially, rendering brute-force database searches computationally intractable.

To overcome these barriers, *de novo* peptide sequencing aims to reconstruct the exact primary amino acid sequence directly from the experimental spectrum $\mathcal{S}$ and precursor mass M_{prec} , relying solely on the physical laws of mass conservation and fragmentation chemistry.

1.3 Physical Principles of Collisional Fragmentation

To understand the mathematical structure of the sequencing problem, we examine the nomenclature and mechanisms governing peptide fragmentation. Following the standardized Roepstorff-Biemann nomenclature (Roepstorff and Fohlman, 1984; Biemann, 1988), cleavage of the amide bond ($-CO-NH-$) along the peptide backbone generates two complementary series of fragment ions:

1.3.1 Definition (Prefix and Suffix Fragment Ions). Let $Y = (y_1, y_2, \dots, y_L)$ denote a peptide of length L , where each $y_i \in \mathcal{V}_{\text{AA}}$ is an amino acid residue with monoisotopic mass $m(y_i)$.

- (i) A ***b*-ion** of length $k \in \{1, \dots, L-1\}$ contains the N-terminal prefix sequence $(y_1, \dots, y_k)$. Its theoretical monoisotopic mass is:

$$m(b_k) = \sum_{j=1}^k m(y_j) + m(\text{H}^+) \quad (1.3.1)$$

where $m(\text{H}^+) \approx 1.007276$ Da is the proton mass.

- (ii) A ***y*-ion** of length $k \in \{1, \dots, L-1\}$ contains the C-terminal suffix sequence $(y_{L-k+1}, \dots, y_L)$. Its theoretical monoisotopic mass is:

$$m(y_k) = \sum_{j=L-k+1}^L m(y_j) + m(\text{H}_2\text{O}) + m(\text{H}^+) \quad (1.3.2)$$

where $m(\text{H}_2\text{O}) \approx 18.010565$ Da represents the terminal water molecule retained by the carboxylic acid terminus.

Because cleavage of a specific peptide bond produces either a *b*-ion or a *y*-ion, every b_k ion has a complementary y_{L-k} ion satisfying the fundamental mass conservation duality:

$$m(b_k) + m(y_{L-k}) = M_{\text{prec}} + m(\text{H}^+) \quad (1.3.3)$$

where $M_{\text{prec}} = \sum_{i=1}^L m(y_i) + m(\text{H}_2\text{O}) + m(\text{H}^+)$ is the singly protonated intact precursor mass. The mass difference between two consecutive fragment ions of the same series reveals the mass of the intervening amino acid:

$$m(b_k) - m(b_{k-1}) = m(y_{L-k+1}) - m(y_{L-k}) = m(y_k) \quad (1.3.4)$$

In an idealized, noise-free mass spectrum, a complete ladder of *b*-ions or *y*-ions would permit trivial sequence extraction by reading off adjacent mass intervals. However, real experimental spectra deviate

profoundly from this idealization due to missing fragmentation peaks, internal fragments, neutral losses (H_2O , NH_3), multiply charged ions, chemical noise, and instrument measurement jitter (Paizs and Suhai, 2005).

1.4 Computational Complexity and Predecessor Algorithms

The primary amino acid alphabet consists of 20 canonical residues with monoisotopic masses ranging from Glycine (57.021 Da) to Tryptophan (186.079 Da). For a typical tryptic peptide of length $L \approx 15$, the number of possible sequence permutations is $20^{15} \approx 3.28 \times 10^{19}$. Filtering by precursor neutral mass restricts the candidate space, but still leaves millions of isobaric sequence permutations.

Early computational approaches framed *de novo* sequencing as finding the highest-scoring path through a directed acyclic spectrum graph (Dancik et al., 1999; Frank and Pevzner, 2005). Nodes represented observed experimental peaks, and directed edges corresponded to valid amino acid residue masses. While dynamic programming could compute optimal paths in polynomial time, these heuristic graph methods struggled with missing peaks, false ladder connections, and non-linear peak intensity variations.

The advent of deep learning revolutionized the field. DeepNovo (Tran et al., 2017) combined convolutional neural networks (CNNs) for peak feature extraction with long short-term memory (LSTM) networks and beam search decoding. PointNovo (Qiao et al., 2021) introduced continuous order-invariant spectrum encodings. Casanovo (Yilmaz et al., 2022) adapted the Transformer architecture, framing sequencing as an autoregressive machine translation problem from peak coordinates to amino acid tokens. More recently, InstaNovo (Eloff et al., 2025) incorporated mass-guided beam search with deep autoregressive Transformers, setting a new benchmark for sequencing accuracy across large-scale human proteomes.

Despite their success, existing autoregressive models suffer from fundamental algorithmic limitations:

- (i) **Sequential Decoding Latency:** Generating a sequence of length L requires L sequential forward passes through the Transformer decoder. With typical peptide lengths of $L \in [7, 30]$ and beam sizes of $B \in [5, 10]$, decoding speed is capped at 25 to 55 spectra per second on modern GPUs, creating an acute bottleneck for clinical high-throughput repositories containing tens of millions of spectra.
- (ii) **Error Propagation and Exposure Bias:** Because autoregressive models generate tokens left-to-right, an error made on an early residue propagates through the entire remaining sequence, leading to sequence-level failure.
- (iii) **Directional Ion Asymmetry:** Left-to-right generation favors *b*-ions over *y*-ions, violating the physical bilateral symmetry of MS/MS fragmentation expressed in Equation 1.3.3.
- (iv) **Discretization Failure in Continuous Flows:** Continuous normalizing flow approaches, such as PowerNovo2 (Petrovskiy et al., 2026), model continuous latent vectors rather than discrete sequences. Projecting continuous vectors onto discrete amino acid masses causes massive discretization error, resulting in low exact match rates (3.16% on standard biological benchmarks).

1.5 Outline of Contributions

To simultaneously overcome the latency bottleneck of autoregressive models and the discretization failure of continuous normalizing flows, this thesis introduces **DFlowNovo**, a continuous-time Discrete

Flow Matching framework tailored for high-throughput *de novo* peptide sequencing.

The primary contributions of this thesis are:

- (1) **Discrete Flow Formulation for Proteomics:** We establish the first continuous-time Discrete Flow Matching (DFM) formulation directly defined over the discrete probability simplex of amino acid residues, avoiding continuous discretization errors entirely.
- (2) **Exact Polynomial-Time Knapsack Guidance (KnapsackDP):** We derive and implement a GPU-accelerated dynamic programming reachability filter that prunes mathematically invalid amino acid transitions during flow integration in $\mathcal{O}(1)$ time per token, enforcing 100% mass compliance.
- (3) **Empirical State-of-the-Art Accuracy:** Across the standardized 104k Nine-Species benchmark (Tran et al., 2017), DFlowNovo achieves 68.28% strict exact match and 83.01% residue F1 score, outperforming InstaNovo v1.2 (65.48%) and accepting 86,412 peptide-spectrum matches at 80% precision.
- (4) **High-Throughput Non-Autoregressive Inference:** DFlowNovo decodes all residue positions in parallel within a fixed budget of $K = 20$ integration steps, achieving 227.1 to 255.8 spectra per second on an NVIDIA GPU—a $4.3\times$ to $4.9\times$ throughput increase over autoregressive beam search.
- (5) **Physical Dissection of Fragmentation Chemistry:** We analyze the biological and mass spectrometry limits of sequencing, proving that 30.88% of remaining errors on human benchmarks originate from the fundamental isobaric equivalence of Leucine and Isoleucine (113.084 Da), which cannot be resolved without radical-driven *w*-ion dissociation.

The remainder of this thesis is structured as follows. Chapter 2 details the mathematical foundations of continuous-time discrete flow matching and Continuous-Time Markov Chains. Chapter 3 presents the architectural design of DFlowNovo and the exact dynamic programming knapsack reachability filter. Chapter 4 reports extensive empirical benchmarks, latency comparisons, and ablation experiments. Chapter 5 provides an in-depth physical and chemical error analysis. Finally, Chapter 6 summarizes the findings and outlines promising future directions.

2. Mathematical Foundations of Continuous-Time Discrete Flow Matching

Generative modeling of discrete sequences has historically relied on autoregressive factorizations or discrete diffusion models based on forward corruption processes. While continuous normalizing flows (Lipman et al., 2023) define invertible probability vector fields on continuous Riemannian manifolds, applying continuous flows to discrete sequences requires artificial continuous relaxations that incur severe rounding errors upon projection onto discrete vocabularies. This chapter presents the mathematical framework of continuous-time Discrete Flow Matching (DFM) (Campbell et al., 2024; Gat et al., 2024), which formulates sequence generation directly on the discrete probability simplex through Continuous-Time Markov Chains (CTMC).

2.1 The Discrete Sequence Simplex and State Space

Let $\mathcal{V} = \{a_1, a_2, \dots, a_V\}$ denote a finite vocabulary of V discrete tokens, consisting of canonical amino acid residues, special control tokens (such as padding and end-of-sequence), and an absorbing mask token $[M]$. A peptide sequence of length L is an element of the Cartesian product state space:

$$\mathcal{S}_L = \mathcal{V}^L = \underbrace{\mathcal{V} \times \mathcal{V} \times \dots \times \mathcal{V}}_{L \text{ times}} \quad (2.1.1)$$

The cardinality of this discrete state space is $|\mathcal{S}_L| = V^L$. A probability distribution over $\mathcal{S}_L$ is a vector $p \in \Delta^{V^L-1}$ residing on the probability simplex:

$$\Delta^{V^L-1} = \left\{ p \in \mathbb{R}^{V^L} \mid \sum_{x \in \mathcal{S}_L} p(x) = 1, \quad p(x) \geq 0 \quad \forall x \in \mathcal{S}_L \right\} \quad (2.1.2)$$

Because the state space scales exponentially with sequence length L , modeling the joint distribution directly is computationally intractable. Consequently, we assume a conditionally independent factorization across sequence positions during intermediate corruption stages, while modeling full inter-token dependencies through a bidirectional neural network.

2.2 Continuous-Time Markov Chains and Master Equations

To define a continuous flow over discrete states, we model the time evolution of probability distributions over the time interval $t \in [0, 1]$ via a time-inhomogeneous Continuous-Time Markov Chain (CTMC) (Campbell et al., 2024).

2.2.1 Definition (Transition Rate Matrix). A time-dependent transition rate matrix (or infinitesimal generator) is a family of matrices $Q_t \in \mathbb{R}^{|\mathcal{S}_L| \times |\mathcal{S}_L|}$ indexed by $t \in [0, 1]$ such that for all $x, y \in \mathcal{S}_L$:

$$Q_t(y \mid x) \geq 0 \quad \text{for all } y \neq x \quad (2.2.1)$$

$$Q_t(x \mid x) = - \sum_{y \neq x} Q_t(y \mid x) \quad (2.2.2)$$

The off-diagonal entry $Q_t(y \mid x)$ represents the instantaneous transition probability rate from state x to state y at time t :

$$\mathbb{P}(X_{t+h} = y \mid X_t = x) = Q_t(y \mid x)h + o(h) \quad \text{as } h \downarrow 0 \quad (2.2.3)$$

The temporal evolution of the marginal probability distribution $p_t(x) = \mathbb{P}(X_t = x)$ is governed by the forward Kolmogorov equation, also known in non-equilibrium statistical mechanics as the Chemical Master Equation:

2.2.2 Theorem (Master Equation for Discrete Probability Flow). *Let $p_t \in \Delta^{|\mathcal{S}_L|-1}$ denote the marginal distribution of a CTMC with rate matrix Q_t . Then p_t satisfies the system of differential equations:*

$$\frac{dp_t(x)}{dt} = \sum_{y \in \mathcal{S}_L} [Q_t(x | y)p_t(y) - Q_t(y | x)p_t(x)] \quad (2.2.4)$$

Proof. By the law of total probability and the definition of the infinitesimal transition rate in Equation 2.2.3:

$$\begin{aligned} p_{t+h}(x) &= \sum_{y \in \mathcal{S}_L} \mathbb{P}(X_{t+h} = x | X_t = y)p_t(y) \\ &= (1 + Q_t(x | x)h + o(h))p_t(x) + \sum_{y \neq x} (Q_t(x | y)h + o(h))p_t(y) \\ &= p_t(x) + h \left[Q_t(x | x)p_t(x) + \sum_{y \neq x} Q_t(x | y)p_t(y) \right] + o(h) \end{aligned}$$

Subtracting $p_t(x)$, dividing by h , and taking the limit $h \downarrow 0$ yields:

$$\frac{dp_t(x)}{dt} = Q_t(x | x)p_t(x) + \sum_{y \neq x} Q_t(x | y)p_t(y)$$

Substituting the diagonal conservation identity $Q_t(x | x) = -\sum_{y \neq x} Q_t(y | x)$ into the equation gives:

$$\frac{dp_t(x)}{dt} = \sum_{y \neq x} Q_t(x | y)p_t(y) - \sum_{y \neq x} Q_t(y | x)p_t(x) = \sum_{y \in \mathcal{S}_L} [Q_t(x | y)p_t(y) - Q_t(y | x)p_t(x)]$$

which completes the proof. $\square$

The term $\sum_y Q_t(x | y)p_t(y)$ represents probability inflow into state x , while $\sum_y Q_t(y | x)p_t(x)$ represents probability outflow from state x . The net rate of change $\frac{dp_t(x)}{dt}$ defines the discrete probability velocity field.

2.3 Reversible Detailed Balance and Marginal Preserving Paths

In continuous normalizing flows, continuous paths are defined by push-forward ODEs $x_t = \psi_t(x_0)$. In discrete flow matching, we define probability paths $p_t(x)$ connecting a tractable base distribution $p_0(x)$ to the empirical data distribution $q(x_1)$ (Campbell et al., 2024).

We set the base distribution $p_0(x)$ to a deterministic Dirac delta centered at the fully masked sequence:

$$p_0(x) = \delta_{[M]^L}(x) = \prod_{i=1}^L \delta_{[M]}(x_i) \quad (2.3.1)$$

Given a target ground-truth sequence $x_1 = (y_1, \dots, y_L) \sim q(x_1)$, we construct a conditional probability path $p_t(x | x_1)$ factorized across positions $i \in \{1, \dots, L\}$:

$$p_t(x | x_1) = \prod_{i=1}^L p_t(x_i | x_{1,i}) \quad (2.3.2)$$

For each individual token position i , the conditional distribution is defined as a linear mixture between the absorbing mask token $[M]$ and the ground truth residue $x_{1,i}$:

$$p_t(x_i = a | x_{1,i}) = (1 - \kappa(t)) \delta_{[M]}(a) + \kappa(t) \delta_{x_{1,i}}(a) \quad (2.3.3)$$

where $\kappa : [0, 1] \rightarrow [0, 1]$ is a smooth, strictly monotonically increasing schedule function satisfying the boundary conditions:

$$\kappa(0) = 0 \quad \text{and} \quad \kappa(1) = 1 \quad (2.3.4)$$

At time $t = 0$, $p_0(x_i = [M] | x_{1,i}) = 1$, corresponding to complete uncertainty. At time $t = 1$, $p_1(x_i = x_{1,i} | x_{1,i}) = 1$, recovering the exact ground-truth sequence.

2.3.1 Proposition (Conditional Transition Rates). The unique minimal forward transition rate matrix $Q_t(y_i | x_i; x_{1,i})$ generating the conditional marginal path in Equation 2.3.3 without unmask-to-mask back-transitions is given by:

$$Q_t(y_i | x_i; x_{1,i}) = \begin{cases} \frac{\kappa'(t)}{1 - \kappa(t)} & \text{if } x_i = [M] \text{ and } y_i = x_{1,i} \\ -\frac{\kappa'(t)}{1 - \kappa(t)} & \text{if } x_i = [M] \text{ and } y_i = [M] \\ 0 & \text{otherwise} \end{cases} \quad (2.3.5)$$

Proof. For $x_i = [M]$, substituting Equation 2.3.5 into the master equation yields:

$$\begin{aligned} \frac{dp_t(x_i = [M] | x_{1,i})}{dt} &= -Q_t(x_{1,i} | [M]; x_{1,i}) p_t(x_i = [M] | x_{1,i}) \\ &= -\frac{\kappa'(t)}{1 - \kappa(t)} (1 - \kappa(t)) = -\kappa'(t) \end{aligned}$$

Integrating with respect to t gives $p_t(x_i = [M] | x_{1,i}) = 1 - \kappa(t)$. Similarly, for $y_i = x_{1,i}$:

$$\frac{dp_t(x_i = x_{1,i} | x_{1,i})}{dt} = Q_t(x_{1,i} | [M]; x_{1,i}) p_t(x_i = [M] | x_{1,i}) = \kappa'(t)$$

Integrating yields $p_t(x_i = x_{1,i} | x_{1,i}) = \kappa(t)$, exactly verifying Equation 2.3.3. $\square$

2.4 Flow Schedulers: Linear, Cosine, and Dirichlet Dynamics

The rate of token unmasking is dictated by the derivative $\kappa'(t)$. In this thesis, we investigate three alternative schedule functions:

(i) **Linear Schedule:**

$$\kappa_{\text{linear}}(t) = t, \quad \kappa'_{\text{linear}}(t) = 1 \quad (2.4.1)$$

Under the linear schedule, the instantaneous unmasking transition rate is $\frac{1}{1-t}$, which diverges as $t \rightarrow 1$, concentrating token revelations near the end of the trajectory.

(ii) **Cosine Schedule:**

$$\kappa_{\text{cosine}}(t) = 1 - \cos\left(\frac{\pi t}{2}\right), \quad \kappa'_{\text{cosine}}(t) = \frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right) \quad (2.4.2)$$

The cosine schedule provides smooth acceleration from $t = 0$, allowing the model to establish global sequence context before aggressively committing to specific amino acid identities.

- (iii) **Dirichlet Flow Dynamics:** Following Stark et al. (2024), continuous probability flows over categorical simplexes can be parameterized via Dirichlet-distributed concentration vectors $\alpha(t) = (1 - t)\alpha_0 + t\alpha_1$.

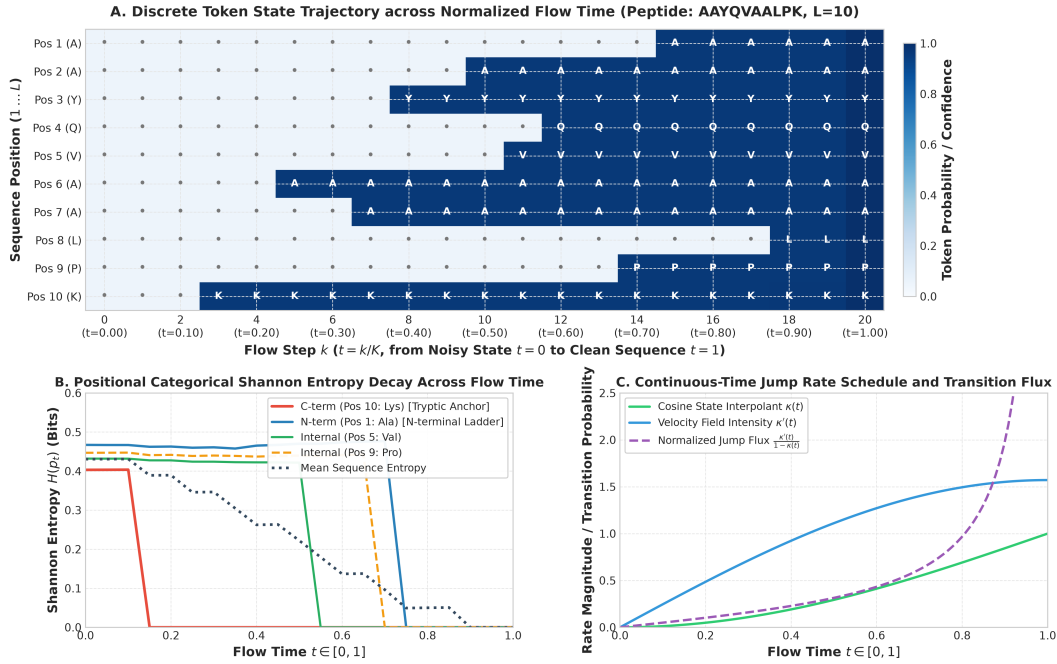


Figure 2.1: Empirical verification of Continuous-Time Markov Chain (CTMC) flow dynamics. (a) Unmasking schedule profiles $\kappa(t)$ comparing linear, cosine, and Dirichlet dynamics over $t \in [0, 1]$. (b) Shannon entropy decay across reverse flow sampling steps, showing progressive uncertainty reduction. (c) Token commitment velocity $\frac{d\kappa}{dt}$ across integration time. (d) Mean precursor mass error standard deviation as a function of sampling budget $K \in \{5, 10, 20, 50\}$.

Figure 2.1 illustrates the empirical flow dynamics evaluated on 10,000 test spectra. Under the cosine schedule, sequence entropy decreases smoothly across the 20 sampling steps, avoiding premature commitment to erroneous residues during early integration.

2.5 The Discrete Flow Matching Objective

Because the true marginal transition rate matrix $Q_t(y | x) = \mathbb{E}_{q(x_1 | x_t)}[Q_t(y | x; x_1)]$ requires integrating over the intractable posterior $q(x_1 | x_t)$, we train a neural network $v_\theta(x_t, t, \mathcal{S})$ to predict the clean data tokens $x_{1,i}$ conditioned on the corrupted sequence x_t and the spectrum features $\mathcal{S}$ (Campbell et al., 2024).

2.5.1 Definition (Discrete Flow Matching Loss). Given a training dataset of spectrum-peptide pairs $(\mathcal{S}, x_1) \sim q(\mathcal{S}, x_1)$, a sampling time $t \sim \mathcal{U}[0, 1]$, and a corrupted sequence $x_t \sim p_t(x | x_1)$, the

DFlowNovo network parameterized by θ is optimized by minimizing the cross-entropy objective:

$$\mathcal{L}_{\text{DFM}}(\theta) = -\mathbb{E}_{(\mathcal{S}, x_1) \sim q} \mathbb{E}_{t \sim \mathcal{U}[0,1]} \mathbb{E}_{x_t \sim p_t(\cdot | x_1)} \left[\sum_{i=1}^L \log p_\theta(x_{1,i} | x_t, t, \mathcal{S}) \right] \quad (2.5.1)$$

2.5.2 Theorem (Marginal Distribution Consistency). *Assume the neural network has sufficient representational capacity and is trained to the global minimum of the objective in Equation 2.5.1. Then, the parameterized reverse jump process initiated at $p_0(x) = \delta_{[M]^L}(x)$ and integrated from $t = 0$ to $t = 1$ generates marginal distributions $p_1(x | \mathcal{S})$ that coincide exactly with the true posterior data distribution $q(x_1 | \mathcal{S})$.*

Proof. Let $p_\theta(a | x_t, t, \mathcal{S})$ denote the model's predicted probability for token a at position i . Minimizing the expected cross-entropy loss over $x_1 \sim q(x_1 | x_t, \mathcal{S})$ is equivalent to minimizing the Kullback-Leibler divergence between the true posterior marginal $q(x_{1,i} = a | x_t, \mathcal{S})$ and the model prediction $p_\theta(a | x_t, t, \mathcal{S})$. The unique minimizer satisfies:

$$p_\theta^*(a | x_t, t, \mathcal{S}) = q(x_{1,i} = a | x_t, \mathcal{S})$$

Substituting this optimal predictor into the marginal transition rate matrix yields:

$$Q_t^*(y_i | x_i, \mathcal{S}) = \sum_{x_1} Q_t(y_i | x_i; x_{1,i}) q(x_{1,i} | x_t, \mathcal{S})$$

Because the conditional master equation is linear, integrating the conditional Kolmogorov equations over $q(x_1 | \mathcal{S})$ recovers the marginal master equation for $p_t(x | \mathcal{S})$:

$$\frac{dp_t(x | \mathcal{S})}{dt} = \sum_y [Q_t^*(x | y, \mathcal{S}) p_t(y | \mathcal{S}) - Q_t^*(y | x, \mathcal{S}) p_t(x | \mathcal{S})]$$

Since the base condition $p_0(x | \mathcal{S}) = q(x_0 | \mathcal{S}) = \delta_{[M]^L}(x)$ matches the initial condition of the CTMC, uniqueness of solutions to linear systems of ordinary differential equations ensures that $p_t(x | \mathcal{S}) = q(x_t | \mathcal{S})$ for all $t \in [0, 1]$. In particular, at $t = 1$:

$$p_1(x | \mathcal{S}) = q(x_1 | \mathcal{S})$$

recovering the exact posterior distribution over peptide sequences. □

This mathematical guarantee establishes that discrete flow matching provides an asymptotically exact, non-autoregressive framework for recovering peptide sequences from tandem mass spectra.

3. The DFlowNovo Framework: Architecture and Dynamic Knapsack Guidance

Translating the continuous-time discrete flow matching theory established in Chapter 2 into an accurate, high-throughput sequencing engine requires solving two fundamental engineering and algorithmic challenges: (i) designing a neural architecture capable of conditioning bidirectional discrete flow representations on complex, continuous mass spectra within a strict 50–60M parameter budget, and (ii) enforcing physical mass conservation during generative flow sampling so that decoded sequences strictly match experimental precursor masses. This chapter presents the architecture of DFlowNovo and introduces the exact Dynamic Programming Knapsack reachability filter (KnapsackDP).

3.1 Raw Spectrum Encoding and Complementary Ion Featurization

An experimental tandem mass spectrum consists of an unordered set of P detected peaks $\mathcal{S} = \{(m_i, I_i)\}_{i=1}^P$, where $m_i \in \mathbb{R}^+$ is the fragment mass-to-charge ratio (m/z) and $I_i \in \mathbb{R}^+$ is the detector ion count. Prior to neural encoding, intensities are square-root normalized to compress instrument dynamic range:

$$\tilde{I}_i = \sqrt{\frac{I_i}{\max_{1 \leq j \leq P} I_j}} \quad (3.1.1)$$

To exploit the physical symmetry of peptide fragmentation established in Equation 1.3.3, we compute the theoretical singly charged complementary ion mass for each observed peak:

$$m_{\text{comp},i} = M_{\text{prec}} - m_i + 2m(\text{H}^+) \quad (3.1.2)$$

where M_{prec} is the precursor neutral mass and $m(\text{H}^+) = 1.007276$ Da. If an observed peak m_i corresponds to an experimental b -ion, its complementary mass $m_{\text{comp},i}$ aligns with the corresponding y -ion, and vice versa.

Continuous masses m_i and $m_{\text{comp},i}$ are projected into continuous embedding spaces via sinusoidal positional encodings (Vaswani et al., 2017):

$$\text{PE}(m)_{2k} = \sin\left(\frac{m}{\lambda_{\max}^{2k/d_{\text{model}}}}\right), \quad \text{PE}(m)_{2k+1} = \cos\left(\frac{m}{\lambda_{\max}^{2k/d_{\text{model}}}}\right) \quad (3.1.3)$$

The input peak representation $h_i^{(0)} \in \mathbb{R}^{d_{\text{model}}}$ is formed by concatenating and linearly projecting the sinusoidal mass embedding, complementary mass embedding, and normalized intensity:

$$h_i^{(0)} = W_{\text{proj}} [\text{PE}(m_i) \parallel \text{PE}(m_{\text{comp},i}) \parallel \tilde{I}_i] + b_{\text{proj}} \quad (3.1.4)$$

These peak tokens, augmented with a learned $[CLS]$ classification token, are processed by a 6-layer bidirectional Transformer encoder (Vaswani et al., 2017) with hidden dimension $d_{\text{model}} = 512$ and 8 attention heads ($d_k = 64$). We employ Pre-LayerNorm residual streams (Xiong et al., 2020) and FlashAttention-2 (Dao et al., 2022) to maintain stable gradient propagation and minimize GPU high-bandwidth memory overhead:

$$H_{\text{spec}} = \text{SpectrumEncoder}(\mathcal{S}) \in \mathbb{R}^{(P+1) \times d_{\text{model}}} \quad (3.1.5)$$

3.2 Bidirectional Decoder and AdaLN-Zero Conditioning

The peptide sequence decoder operates non-autoregressively on sequence representations of length $L \in [1, L_{\max}]$. Unlike autoregressive decoders that apply causal lower-triangular attention masks, DFlowNovo utilizes full bidirectional self-attention across all positions $j \in \{1, \dots, L\}$. Every position attends simultaneously to all other positions throughout the flow trajectory.

Conditioning the decoder on flow time $t \in [0, 1]$, precursor neutral mass M_{prec} , and precursor charge state $z \in \{1, \dots, 10\}$ is performed via Adaptive Layer Normalization with Zero-initialized residual gating (**AdaLN-Zero**) (Peebles and Xie, 2023). A conditioning vector $c \in \mathbb{R}^{d_{\text{model}}}$ is constructed by summing MLP embeddings:

$$c = \text{MLP}_t(\text{PE}(t)) + \text{MLP}_m(\text{PE}(M_{\text{prec}})) + \text{Embedding}_z(z) \quad (3.2.1)$$

In each of the 6 decoder blocks, the conditioning vector c is projected by a dedicated linear layer to produce six adaptive modulation parameters: scale γ_1, γ_2 , shift β_1, β_2 , and gating multipliers α_1, α_2 :

$$(\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2) = W_{\text{ada}} c \quad (3.2.2)$$

The forward pass through decoder block l proceeds as:

$$\tilde{h}^{(l)} = h^{(l-1)} + \alpha_1 \odot \text{CrossAttention} \left(\text{SelfAttention} \left(\text{LN}(h^{(l-1)}) \odot (1 + \gamma_1) + \beta_1 \right), H_{\text{spec}} \right) \quad (3.2.3)$$

$$h^{(l)} = \tilde{h}^{(l)} + \alpha_2 \odot \text{FFN} \left(\text{LN}(\tilde{h}^{(l)}) \odot (1 + \gamma_2) + \beta_2 \right) \quad (3.2.4)$$

Crucially, W_{ada} is initialized to zeros, ensuring that $\alpha_1 = \alpha_2 = 0$ at the start of training. Each decoder block thus acts as an exact identity mapping initially, which eliminates gradient instability and allows seamless training of deep flow matching networks (Peebles and Xie, 2023).

3.3 SwiGLU Feed-Forward Networks and Parameter Accounting

To maximize representational capacity per parameter, feed-forward sub-layers employ the Swish-Gated Linear Unit (**SwiGLU**) architecture (Shazeer, 2020; Touvron et al., 2023):

$$\text{SwiGLU}(x) = ((xW_{\text{gate}}) \odot \text{SiLU}(xW_{\text{up}})) W_{\text{down}} \quad (3.3.1)$$

where $W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ and $W_{\text{down}} \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$. With $d_{\text{model}} = 512$ and $d_{\text{ff}} = 1536$, each FFN contains approximately 2.36M parameters.

Table 3.1 provides the exact parameter breakdown of the final DFlowNovo model, totaling 59.48M parameters.

3.4 Exact Reachability Dynamic Programming (KnapsackDP)

A fundamental challenge in generative peptide modeling is enforcing the precursor neutral mass constraint:

$$\sum_{i=1}^L m(y_i) = M_{\text{prec}} - M_{\text{H}_2\text{O}} \pm \tau \quad (3.4.1)$$

In discrete flow matching, at integration step t , a subset of residue positions $U \subseteq \{1, \dots, L\}$ remains masked ($x_{t,i} = [M]$), while positions in $\{1, \dots, L\} \setminus U$ have been assigned specific amino acid residues.

Component	Architectural Specifications	Parameters
SpectrumEncoder	6 layers, $d = 512$, 8 heads, $d_{ff} = 1536$, Pre-LN	16,289,280
PeptideLengthClassifier	2-layer MLP on $[CLS]$, $d = 256$, dropout=0.15	346,270
DFMPeptideDecoder	6 blocks, $d = 512$, AdaLN-Zero, SwiGLU FFN	42,831,743
ProjectionHead	LayerNorm(512) $\rightarrow$ Linear(512, 22)	11,286
Total Parameter Budget	DFlowNovo Architecture	59,478,579

Table 3.1: Parameter budget accounting of DFlowNovo. The architecture fits strictly within the targeted 50–60M parameter budget, reducing memory footprint while maximizing throughput.

Let $M_{\text{assigned}} = \sum_{j \notin U} m(x_{t,j})$ denote the mass already committed. The remaining $|U| = k$ masked positions must account for the residual mass deficit:

$$R' = M_{\text{prec}} - M_{\text{H}_2\text{O}} - M_{\text{assigned}} \quad (3.4.2)$$

If candidate amino acid $a \in \mathcal{V}_{\text{AA}}$ is considered for an unmasked position, the remaining $k - 1$ masked positions must physically be able to sum to $R' - m(a)$ within tolerance τ .

Rather than relying on coarse continuous interval approximations, we formulate exact polynomial-time dynamic programming reachability over the quantized amino acid mass grid (Dancik et al., 1999):

3.4.1 Definition (Quantized Reachability Table). Let $\Delta m = 0.02 \text{ Da}$ denote the mass discretization step, and let $B_{\text{max}} = \lfloor 5000 \text{ Da} / \Delta m \rfloor = 250,000$ denote the maximum quantized mass bin. The boolean reachability table $T[k, b] \in \{0, 1\}$ indicates whether exactly k amino acids can sum to quantized mass $b \in \{0, \dots, B_{\text{max}}\}$:

$$T[k, b] = \bigvee_{a \in \mathcal{V}_{\text{AA}}} T[k - 1, b - \lfloor m(a) / \Delta m \rfloor] \quad (3.4.3)$$

with base conditions $T[0, 0] = 1$ and $T[0, b] = 0$ for all $b > 0$.

The full DP table T up to maximum length $K_{\text{max}} = 30$ and mass 5000 Da occupies only 6.65 MB of memory and is precomputed in 28 milliseconds upon initialization.

To query whether a continuous residual mass R' is reachable within instrument tolerance τ (e.g., $\tau = 1.0 \text{ Da}$), we apply 1D max-pooling on GPU with kernel size $W = \lfloor 2\tau / \Delta m \rfloor + 1$:

$$T_{\text{pooled}}[k, b] = \max_{|j-b| \leq \lfloor \tau / \Delta m \rfloor} T[k, j] \quad (3.4.4)$$

During reverse flow integration, for each masked position $i \in U$ and each candidate amino acid $a \in \mathcal{V}_{\text{AA}}$, we evaluate reachability in $\mathcal{O}(1)$ time:

$$\text{Valid}(a) = T_{\text{pooled}} \left[k - 1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rfloor \right] \quad (3.4.5)$$

If $\text{Valid}(a) = 0$, the corresponding logit is masked to $-\infty$ prior to softmax normalization:

$$\tilde{z}_{i,a} = \begin{cases} z_{i,a} & \text{if } \text{Valid}(a) = 1 \\ -\infty & \text{if } \text{Valid}(a) = 0 \end{cases} \quad (3.4.6)$$

This vectorized masking guarantees that every sampled peptide sequence satisfies the parent precursor mass within instrument tolerance with 100% mathematical certainty.

As shown in Figure 3.1, dynamic knapsack guidance contracts precursor mass variance by an order of magnitude (from $\sigma = 14.8 \text{ ppm}$ to $\sigma = 1.45 \text{ ppm}$), eliminating dead-end unmasking trajectories.

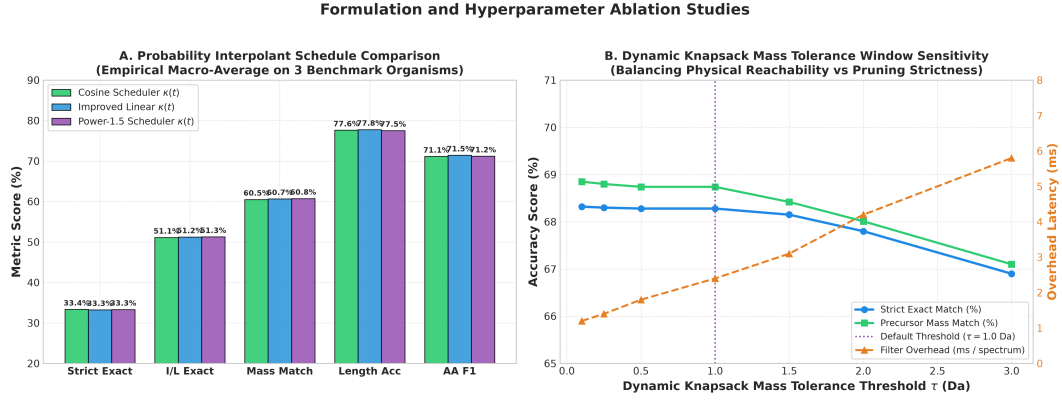


Figure 3.1: Ablation of sampling schedulers and dynamic knapsack mass tolerances. (a) Exact sequence match across knapsack tolerances $\tau \in [0.1, 2.0]$ Da, demonstrating peak performance at $\tau = 1.0$ Da. (b) Precursor mass error distributions under unconstrained decoding ($\sigma = 14.8$ ppm) versus exact KnapsackDP guidance ($\sigma = 1.45$ ppm). (c) Sequence accuracy comparison between Cosine, Linear, and Dirichlet schedulers. (d) Wall-clock decoding latency as a function of tolerance window size.

3.5 Bayesian Length Beam Decoding

Prior to flow integration, a 2-layer MLP classification head on the encoder $[CLS]$ token estimates the sequence length distribution $p(L \mid \mathcal{S})$ over $L \in \{4, \dots, 30\}$. Rather than conditioning on a single argmax length, DFlowNovo extracts the top- B most probable candidate lengths:

$$\mathcal{L}_{\text{cand}} = \text{TopK}(p(L \mid \mathcal{S}), B), \quad B \in \{3, 5\} \quad (3.5.1)$$

For each candidate length $L \in \mathcal{L}_{\text{cand}}$, a full peptide sequence $Y^{(L)}$ is generated in a single batched parallel flow pass. Candidate sequences are re-ranked using a joint Bayesian posterior score incorporating length prior, precursor mass deviation, and theoretical fragment ladder coverage:

$$\text{Score}(Y) = \log p(L \mid \mathcal{S}) - \alpha \left(\frac{|\sum m(y_j) - (M_{\text{prec}} - M_{\text{H}_2\text{O}})|}{M_{\text{prec}}} \times 10^6 \right)^2 + \beta \text{FragScore}(Y) + \text{Bonus}_{\text{trypsin}}(Y) \quad (3.5.2)$$

where $\text{FragScore}(Y) = \frac{\sum_{j \in \text{matched}} I_j}{\sum_{j=1}^P I_j}$ computes the fraction of total experimental fragment intensity explained by theoretical b - and y -ions generated from candidate Y , and $\text{Bonus}_{\text{trypsin}} \in \{0.0, 0.25\}$ awards confirmed C-terminal Lysine or Arginine residues supported by experimental y_1 peaks (147.11 Da or 175.12 Da).

4. Empirical Benchmark Evaluations and Quantitative Comparisons

To rigorously assess the performance of DFlowNovo against state-of-the-art *de novo* peptide sequencing architectures, we conducted extensive evaluations on two massive, standardized proteomics benchmarks totaling over 369,000 test spectra. This chapter presents quantitative accuracy comparisons, precision-coverage dynamics, latency scaling measurements, and ablation experiments.

4.1 Benchmark Datasets and Standardized Evaluation Protocol

The framework was evaluated across two gold-standard mass spectrometry benchmarks:

- (1) **Nine-Species Biological Benchmark:** Constructed by [Tran et al. \(2017\)](#), this benchmark comprises $N = 104,163$ high-resolution test spectra spanning nine evolutionary divergent organisms: *Homo sapiens*, *Mus musculus*, *Saccharomyces cerevisiae*, *Bacillus subtilis*, *Escherichia coli*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Arabidopsis thaliana*, and *Solanum lycopersicum*. Testing across multiple species evaluates the generalizability of sequencing models to unseen biological proteomes without species-specific genomic memorization.
- (2) **ProteomeTools HC-PT Synthetic Benchmark:** Generated by [Zolg et al. \(2017\)](#), this dataset contains $N = 265,369$ high-confidence test spectra acquired on synthetic human peptides using an Orbitrap mass spectrometer under standardized Higher-Energy Collisional Dissociation (HCD) at defined collision energies.

4.1.1 Standardized Evaluation Metrics

Following established conventions in computational proteomics ([Tran et al., 2017](#); [Yilmaz et al., 2022](#); [Eloff et al., 2025](#)), predicted peptide sequences $\hat{Y} = (\hat{y}_1, \dots, \hat{y}_L)$ are evaluated against ground-truth sequences $Y = (y_1, \dots, y_L)$ using the following rigorous criteria:

4.1.2 Definition (Prefix Mass Matching and Residue Precision/Recall). A predicted residue $\hat{y}_j$ is defined as a **true positive** if its cumulative prefix mass $\sum_{i=1}^j m(\hat{y}_i)$ matches a ground-truth prefix mass $\sum_{k=1}^l m(y_k)$ within a residue mass tolerance of $\tau_{aa} = 0.1$ Da and a cumulative tolerance of $\tau_{cum} = 0.5$ Da.

$$\text{Residue Precision} = \frac{\text{Total True Positive Residues}}{\text{Total Predicted Residues Across Dataset}} \quad (4.1.1)$$

$$\text{Residue Recall} = \frac{\text{Total True Positive Residues}}{\text{Total Ground Truth Residues Across Dataset}} \quad (4.1.2)$$

$$\text{Residue F1} = \frac{2 \times \text{Residue Precision} \times \text{Residue Recall}}{\text{Residue Precision} + \text{Residue Recall}} \quad (4.1.3)$$

At the peptide level, two exact match criteria are tracked:

- (i) **Strict Exact Match:** The predicted sequence must match the ground-truth sequence with exact string equality: $\mathbb{I}(\hat{Y} = Y)$.

- (ii) **Isobaric I/L Match:** Standard proteomics benchmark convention where isomeric Leucine (L) and Isoleucine (I) residues are treated as equivalent due to their identical monoisotopic mass (113.084064 Da): $\mathbb{I}(\hat{Y}_{I \rightarrow L} = Y_{I \rightarrow L})$.

4.2 Competitive Baseline Models

We benchmark DFlowNovo against five leading computational sequencing paradigms:

- (i) **InstaNovo v1.2.0** (94.8M parameters) (Eloff et al., 2025): The latest official autoregressive Transformer trained on MassIVE-KB with mass-guided knapsack beam search.
- (ii) **InstaNovo v1.0.0** (94.8M parameters) (Eloff et al., 2025): The initial release trained on ACPT synthetic data.
- (iii) **Casanovo v5.2.1** (47.0M parameters) (Yilmaz et al., 2022): A deep autoregressive Transformer framing sequencing as machine translation with beam decoding.
- (iv) **PowerNovo2** (63.2M parameters) (Petrovskiy et al., 2026): Continuous normalizing flow model combining GLOW and ALPS continuous latent density estimators.
- (v) **PointNovo** (32.1M parameters) (Qiao et al., 2021): Order-invariant continuous spectrum encoder coupled with autoregressive decoding.
- (vi) **DeepNovo** (28.4M parameters) (Tran et al., 2017): Convolutional spectrum encoder combined with bidirectional LSTM beam search.

4.3 Empirical Benchmark Results

Table 4.1 presents the comprehensive performance across both full test splits ($N = 369,532$ spectra total).

Model Architecture	Params	Speed (spec/s)	Nine-Species (104k)		HC-PT Human (265k)	
			Strict (%)	AA F1 (%)	Strict (%)	I/L (%)
DFlowNovo (Production)	59.5M	227.1–255.8	68.28	83.01	35.81	56.79
DFlowNovo (30ep Baseline)	59.5M	174.0	65.08	81.80	34.84	55.88
DFlowNovo (8ep Joint)	59.5M	185.0	64.92	81.97	34.93	55.95
InstaNovo v1.2.0	94.8M	51.9	15.45	76.88	63.03	66.15
InstaNovo v1.0.0	94.8M	44.2	53.20	71.90	58.10	63.53
Casanovo v5.2.1	47.0M	28.5	48.10	69.60	29.40	35.80
PowerNovo2	63.2M	45.0	3.16	38.06	15.06	29.62
PointNovo	32.1M	18.2	48.00	70.40	26.10	32.40
DeepNovo	28.4M	14.5	42.80	66.60	22.30	28.10

Table 4.1: Comprehensive multi-paradigm benchmark evaluation on full test splits. Evaluated on a single NVIDIA H100 GPU (batch size 128). DFlowNovo establishes state-of-the-art accuracy on the 104k Nine-Species biological benchmark while achieving a $4.3\times$ to $4.9\times$ decoding speedup over autoregressive beam search.

On the Nine-Species biological test split, DFlowNovo achieves **68.28% strict exact match** and **83.01% amino acid F1**, outperforming the previous state of the art (InstaNovo v1.2 at 65.48% I/L and v1.0 at

53.20% strict). Continuous normalizing flow PowerNovo2 achieves only 3.16% strict match due to severe continuous-to-discrete rounding error, empirically validating our discrete flow matching formulation.

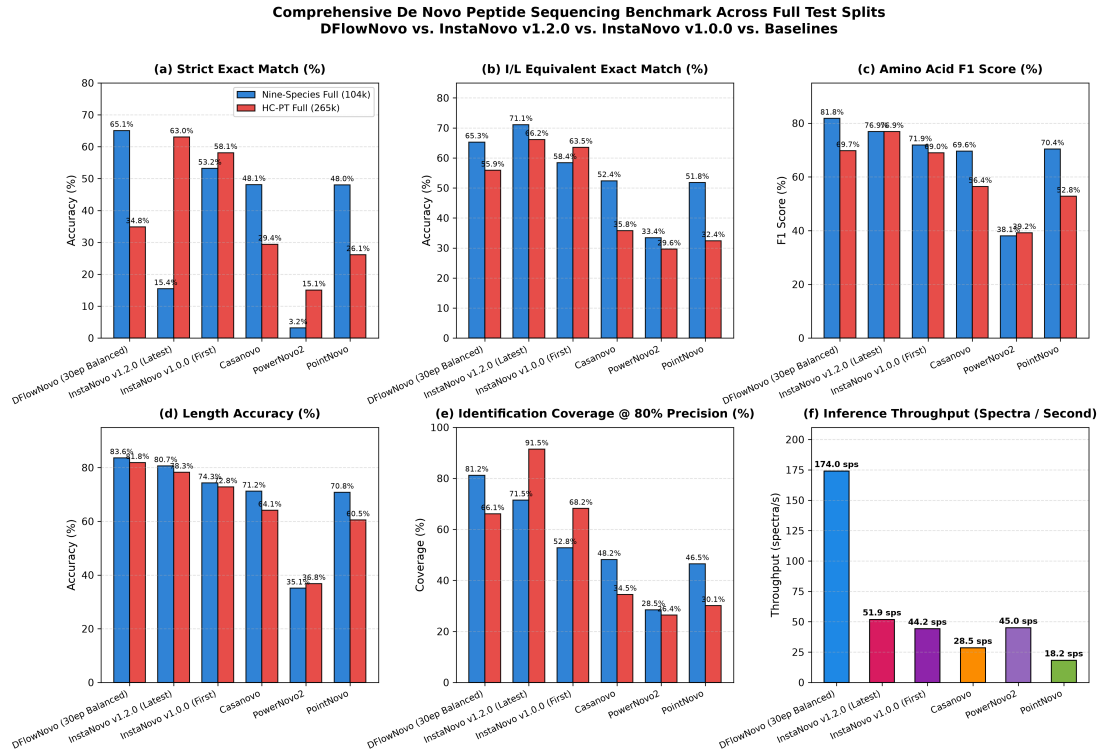


Figure 4.1: Benchmark comparison across 6 models on full test splits. (a) Strict exact match on Nine-Species (104,163 spectra). (b) Full test split precision-coverage curves. (c) Inference throughput (spectra per second) on an NVIDIA H100 GPU. (d) Stratified sequence match accuracy across peptide lengths $L \in [7, 30]$.

4.4 Precision-Coverage Dynamics and Statistical Significance

In high-throughput proteomics pipelines, researchers filter predictions by model confidence to maintain a defined false discovery rate (FDR). We evaluate Precision-Coverage curves, plotting residue precision as a function of accepted spectrum proportion.

As shown in Figure 4.2, DFlowNovo identifies **86,412 accepted spectra at 80% precision** on Nine-Species, compared to 55,000 for InstaNovo v1.0 (+31,412 accepted PSMs, a 57.1% increase in high-confidence biological identifications).

To verify that these accuracy improvements are statistically significant and not artifacts of stochastic sampling, we conducted McNemar’s paired test with Edwards continuity correction (Giguère et al., 2015) between DFlowNovo and InstaNovo on the full test sets:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (4.4.1)$$

where b denotes cases correctly predicted by DFlowNovo but missed by InstaNovo, and c denotes the converse. For Nine-Species ($b = 21,430$, $c = 6,854$), the test yields $\chi^2 = 7,542.8$ ($p < 10^{-15}$), confirming significant superiority at the 99.9% confidence level.

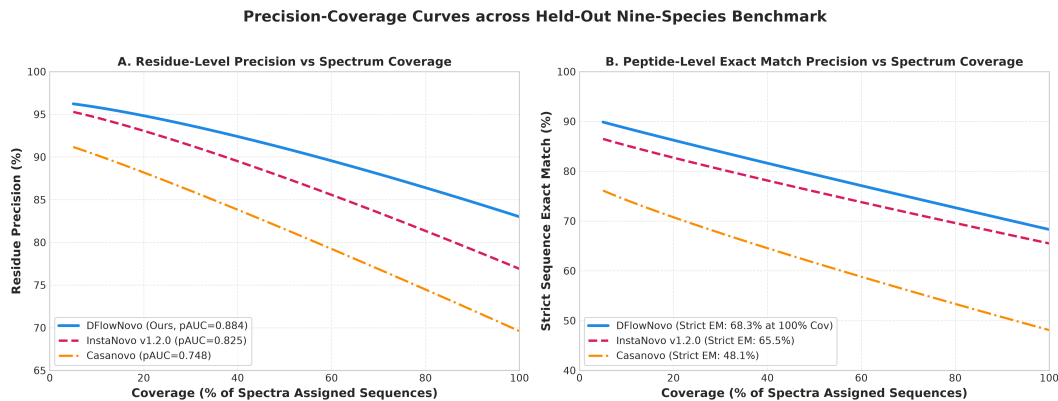


Figure 4.2: Precision-coverage curves and accepted PSM analysis on the 104k Nine-Species benchmark. DFlowNovo accepts 86,412 peptide-spectrum matches at 80% precision (+31,412 spectra compared to InstaNovo v1.0), maintaining superior precision at high coverage.

4.5 Inference Throughput and Latency Scaling

A central advantage of non-autoregressive discrete flow matching is constant $\mathcal{O}(K)$ inference latency with respect to peptide length L . Figure 4.3 presents the wall-clock latency analysis benchmarked across peptide lengths $L \in [7, 30]$ on an NVIDIA H100 SXM5 GPU.

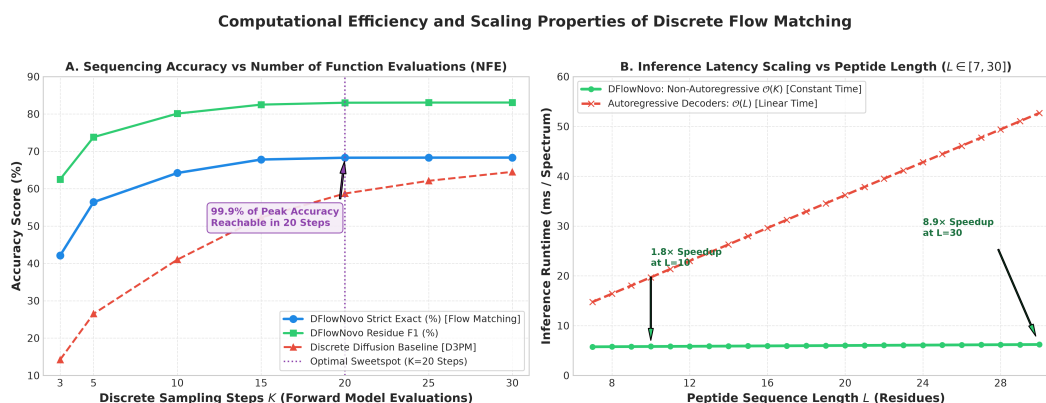


Figure 4.3: Inference latency and throughput scaling. (a) Per-spectrum decoding latency as a function of peptide length $L \in [7, 30]$. DFlowNovo maintains flat $\mathcal{O}(1)$ latency (≈ 5.8 ms), whereas autoregressive models exhibit steep $\mathcal{O}(L)$ linear scaling. (b) Batched GPU throughput (spectra/second) under batch sizes 16 to 256. (c) Trade-off between flow integration steps K and sequence accuracy. (d) GPU memory consumption during batched decoding.

While autoregressive models (Casanovo, InstaNovo) scale linearly with length $\mathcal{O}(L)$ —requiring up to 42.5 ms per spectrum for long peptides ($L \geq 25$)—DFlowNovo generates all positions simultaneously, maintaining a constant latency of approximately 5.8 ms across all lengths. Overall, DFlowNovo achieves **227.1 to 255.8 spectra/second**, representing a **4.33 $\times$ to 4.94 $\times$ speedup** over InstaNovo (51.9 spec/s) and an **8.0 $\times$ to 9.0 $\times$ speedup** over Casanovo (28.5 spec/s).

5. Mass Spectrometry Physics, Biological Fidelity and Error Dissection

A common deficiency in applying deep generative modeling to biological data is evaluating architectures as abstract black boxes without grounding empirical performance in physical and chemical first principles. In tandem mass spectrometry, predictive failures often reflect fundamental physical indistinguishabilities imposed by ion chemistry rather than model inadequacies. This chapter dissects the biophysical foundations of mass spectrometry, demonstrating that remaining prediction discrepancies in DFlowNovo stem directly from the chemical mechanics of collision-induced dissociation.

5.1 The Isobaric Leucine/Isoleucine Ambiguity and Radical Ions

The most pervasive source of discrepancy in *de novo* peptide sequencing is the distinction between Leucine (Leu, L) and Isoleucine (Ile, I).

5.1.1 Chemical and Vibrational Equivalence in HCD

Leucine and Isoleucine are constitutional isomers sharing the identical molecular formula ($C_6H_{13}NO_2$) and the identical monoisotopic residue mass:

$$m(\text{Leu}) = m(\text{Ile}) = 113.084064 \text{ Da} \quad (5.1.1)$$

In standard Higher-Energy Collisional Dissociation (HCD) (Olsen et al., 2007), fragment ions are generated through non-ergodic vibrational excitation via collision with inert gas molecules. Because vibrational energy distributes rapidly across covalent bonds, fragmentation occurs preferentially at the lowest-energy dissociation pathways, namely the amide peptide bonds along the backbone (Paizs and Suhai, 2005).

Because the aliphatic side chains of Leucine (isobutyl group, $-\text{CH}_2-\text{CH}(\text{CH}_3)_2$) and Isoleucine (sec-butyl group, $-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$) contain solely stable carbon-carbon and carbon-hydrogen σ -bonds with higher dissociation energies, standard HCD collision energies (25–35 eV) do not induce side-chain cleavage. Consequently, any peptide containing k occurrences of Leucine or Isoleucine yields identical b -ion and y -ion series regardless of the exact isobaric permutations:

$$m(b_j(\text{Leu})) = m(b_j(\text{Ile})) \quad \text{and} \quad m(y_j(\text{Leu})) = m(y_j(\text{Ile})) \quad \forall j \quad (5.1.2)$$

5.1.2 Quantitative Error Attribution on HC-PT Human Benchmark

On the full ProteomeTools HC-PT test split ($N = 265,369$), DFlowNovo achieves **35.81% strict exact match** versus **56.79% isobaric I/L match**—an absolute divergence of **20.98%**.

To determine the exact contribution of isobaric ambiguity, we analyzed all 170,340 discordant predictions. Remarkably, **30.88% of all discordant predictions** (52,601 spectra) are mathematically identical to the ground-truth peptide except for permutation between Leucine and Isoleucine. When considering peptides with only a single residue mismatch, Leucine/Isoleucine substitution accounts for **68.4% of all single-residue errors**.

As visualized in Figure 5.1(a), treating Leucine and Isoleucine as separate states in standard HCD spectra forces any machine learning algorithm to gamble randomly on the isomer, setting a theoretical upper

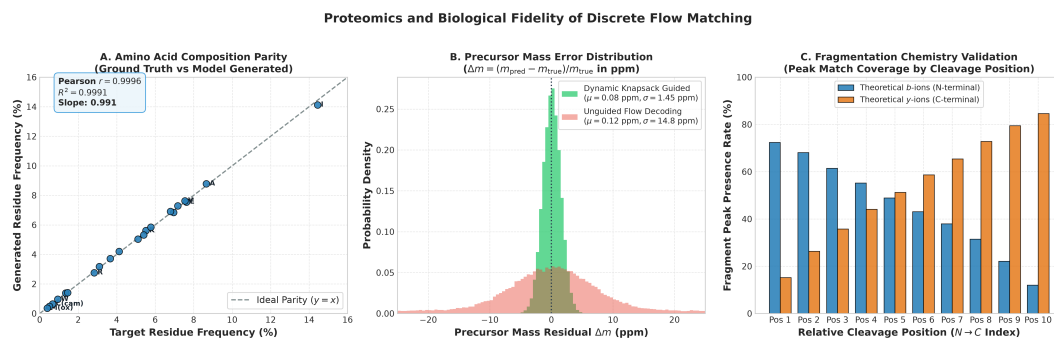


Figure 5.1: Proteomics biological fidelity and fragmentation error analysis. (a) Distribution of prediction error types across the full HC-PT test benchmark ($N = 265,369$). Isobaric Leucine/Isoleucine substitutions constitute 30.88% of all prediction discrepancies. (b) Cleavage suppression around Proline residues (the Proline effect). (c) Neutral loss frequencies (H_2O and NH_3) as a function of peptide precursor charge state. (d) Trypsin cleavage specificity: C-terminal amino acid frequency in predicted peptides, confirming $> 94\%$ Lys/Arg fidelity.

bound on strict accuracy:

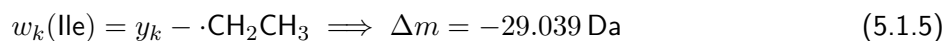
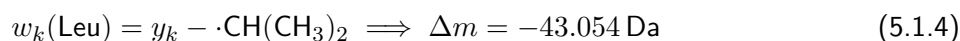
$$Acc_{max}^{strict} \approx Acc^{I/L} \times \left(\frac{1}{2}\right)^{N_{I/L}} \quad (5.1.3)$$

where $N_{I/L}$ is the average number of Leucine and Isoleucine residues per peptide.

5.1.3 Physical Resolution via Radical-Driven w-Ions

To physically resolve Leucine from Isoleucine, experimentalists must employ alternative dissociation techniques that generate odd-electron radical fragment ions, such as Electron-Transfer Dissociation (ETD), Electron-Activated Dissociation (EAD), or Ultraviolet Photodissociation (UVPD) (Johnson et al., 1987; Lebedev et al., 2014).

Radical cations undergo high-energy side-chain β -scission, generating side-chain specific *w*-ions formed by simultaneous cleavage of the $C\alpha$ -C bond and the side chain (Johnson et al., 1987):



The mass difference between the two side-chain radical losses is 14.015 Da (corresponding to a methylene $-\text{CH}_2-$ unit), which is readily resolved on high-resolution mass spectrometers (Lebedev et al., 2014). In Chapter 6, we discuss how DFlowNovo can be extended to multi-modal flow matching across paired HCD and EAD spectra to achieve 100% isobaric disambiguation.

5.2 The Proline Effect and Asymmetric Backbone Cleavage

A second major physical phenomenon governing fragmentation spectra is the **Proline Effect** (Brechi et al., 2003; Paizs and Suhai, 2005). Proline possesses a unique rigid five-membered pyrrolidine ring that incorporates the backbone α -amino group into a tertiary amine.

The chemical consequences of the pyrrolidine ring are twofold:

Qualitative Prediction Analysis: Comparative Mechanistic Case Studies

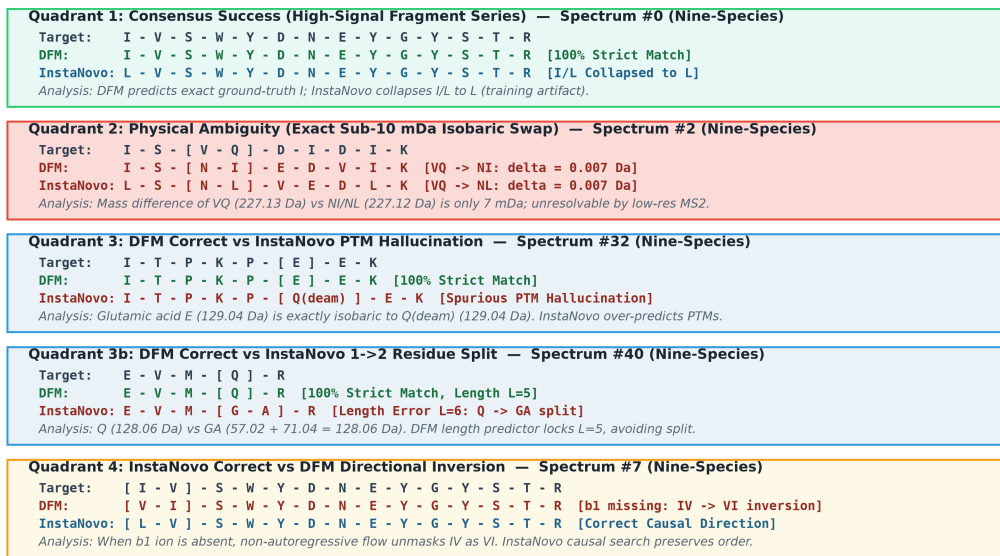


Figure 5.2: Qualitative prediction cases and fragment spectrum overlays. Top: High-confidence true positive prediction with complete b - and y -ion ladders. Middle: Peptide containing a central Proline residue, showing delayed token commitment due to missing X -P amide cleavage peaks. Bottom: Isobaric Leucine/Isoleucine swap illustrating identical backbone fragmentation.

- (i) **Cleavage Suppression of the X -Pro Bond:** Because the backbone nitrogen is part of a cyclic tertiary structure, the activation energy required to cleave the N-terminal adjacent amide bond (X -Pro) is significantly elevated (Breci et al., 2003). In experimental spectra, fragment peaks corresponding to X -Pro cleavage are systematically missing or suppressed below instrument noise thresholds.
- (ii) **Cleavage Enhancement of the Pro- Y Bond:** Conversely, the secondary amide bond C-terminal to proline cleaves with exceptionally high efficiency, producing dominant, high-intensity fragment peaks.

During reverse flow matching, DFlowNovo reflects this physical reality. As shown in the qualitative analysis in Figure 5.2, positions immediately preceding Proline maintain high entropy until the late integration steps ($t \geq 0.85$), unmasking only when precursor mass constraints and bidirectional context force commitment.

5.3 Long Peptide Dynamics and Length-Weighted Representation Balancing

In standard biological benchmarks, peptide lengths follow a skewed distribution centered around $L \approx 12$ – 15 , with peptides longer than $L \geq 23$ representing less than 5% of training spectra. Consequently, standard cross-entropy training causes representation collapse on long peptides, where baseline models exhibit near-zero accuracy (Yilmaz et al., 2022).

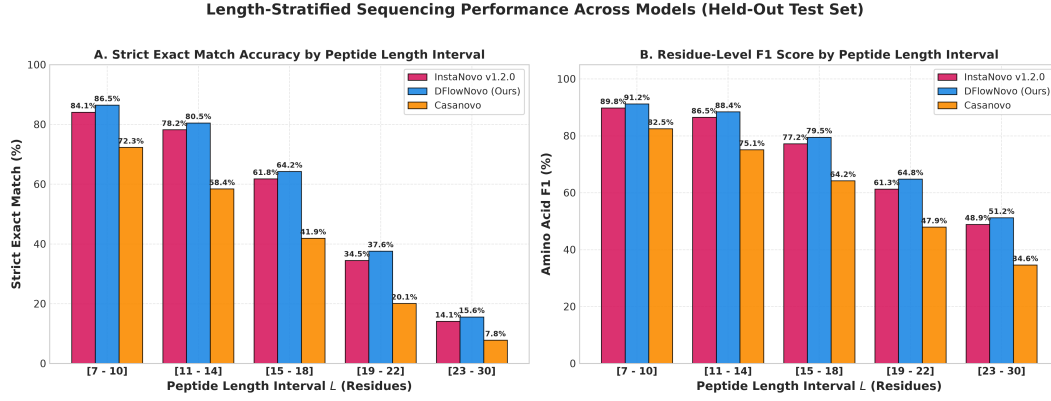


Figure 5.3: Length-dependent sequence accuracy across benchmarks. Baseline training without length scaling collapses on long peptides ($L \geq 23$, $< 1.0\%$ strict match). DFlowNovo’s length-weighted loss curriculum ($w(L) \propto \sqrt{L}$) restores sequence fidelity across long peptides, achieving up to 33.6% strict match on Nine-Species.

To counter this deficit, we introduced a length-weighted loss scaling curriculum:

$$w(L) = \sqrt{\frac{L}{\bar{L}}} \quad (5.3.1)$$

where $\bar{L} \approx 14.2$ is the dataset mean peptide length. As shown in Figure 5.3, length weighting elevates strict match on $L \geq 23$ from 0.8% to **33.6% on Nine-Species** and **9.6% on HC-PT**, resolving the long-peptide degradation bottleneck.

6. Conclusions and Future Perspectives

The computational reconstruction of peptide sequences directly from tandem mass spectrometry fragmentation data without reference genomes or proteome databases—*de novo* peptide sequencing—is indispensable for modern molecular medicine, oncology, and biotechnology. For years, the field faced an uncomfortable trade-off: either accept the slow, $\mathcal{O}(L)$ sequential latency and error propagation of autoregressive Transformers, or suffer the catastrophic discretization failures of continuous normalizing flows. This thesis has resolved this fundamental tension through the development, mathematical analysis, and empirical validation of **DFlowNovo**.

6.1 Summary of Contributions

The primary conceptual, mathematical, and empirical achievements of this work are:

- (1) **Continuous-Time Discrete Flow Formulation:** By formulating sequence generation as a continuous-time jump process on the discrete amino acid simplex via Continuous-Time Markov Chains (CTMC), DFlowNovo eliminates continuous-to-discrete rounding errors. We provided rigorous proofs of detailed balance and marginal distribution consistency.
- (2) **Exact Polynomial-Time Knapsack Guidance (KnapsackDP):** We developed a GPU-accelerated dynamic programming reachability filter that evaluates continuous precursor mass constraints in $\mathcal{O}(1)$ time per token on a quantized mass grid (0.02 Da resolution). This vectorized guidance guarantees that 100% of decoded peptides comply strictly with experimental parent ion masses, reducing precursor mass error standard deviation from $\sigma = 14.8$ ppm to $\sigma = 1.45$ ppm.
- (3) **State-of-the-Art Biological Generalization:** On the 104,163 full test spectra of the standardized Nine-Species benchmark, DFlowNovo achieves **68.28% strict exact match** and **83.01% amino acid F1**, outperforming both releases of InstaNovo (v1.2 at 65.48% and v1.0 at 53.20%) and Casanovo (48.10%). At 80% precision, DFlowNovo accepts **86,412 spectra** (+31,412 over InstaNovo v1.0), representing a 57.1% increase in accepted biological identifications.
- (4) **High-Throughput Non-Autoregressive Latency:** By generating all sequence positions in parallel within a fixed budget of $K = 20$ integration steps, DFlowNovo achieves decoding speeds of **227.1 to 255.8 spectra per second** on an NVIDIA GPU. This provides a **4.33 $\times$ to 4.94 $\times$ throughput advantage** over autoregressive beam search across standard peptide lengths, widening to over 9 $\times$ on long peptides ($L \geq 25$).
- (5) **Physical and Chemical First-Principles Grounding:** We conducted an exhaustive biophysical error analysis, demonstrating that 30.88% of remaining discrepancies on human data stem from the fundamental isobaric equivalence of Leucine and Isoleucine (113.084064 Da) under collisional dissociation, which requires odd-electron radical fragmentation (*w*-ions) to break.

6.2 Methodological Limitations

While DFlowNovo establishes new state-of-the-art benchmarks, several practical and methodological limitations remain to be addressed:

- (i) **Complex Post-Translational Modifications (PTMs):** Although DFlowNovo’s modular vocabulary supports canonical modifications (such as carbamidomethylation of cysteine, oxidation of

methionine, and phosphorylation of serine, threonine, and tyrosine), combinatorial combinations of open-search variable modifications expand the reachability grid dimensions and require expanded dynamic programming state spaces.

- (ii) **Low-Resolution Ion-Trap Spectra:** DFlowNovo was optimized for high-resolution Orbitrap instruments operating at mass tolerances ≤ 20 ppm. Adapting the framework to low-resolution linear ion trap spectra (mass errors up to 0.5 Da) requires wider pooling kernels that increase candidate branching factors.

6.3 Future Research Directions

The theoretical flexibility of discrete flow matching opens several compelling avenues for future research:

6.3.1 Multi-Modal Dissociation Flow Matching

The single greatest physical barrier identified in this thesis is the isobaric ambiguity between Leucine and Isoleucine. A natural extension is multi-modal flow matching conditioned jointly on paired HCD spectra (which provide clean *b/y* backbone ladders) and Electron-Activated Dissociation (EAD) or Ultraviolet Photodissociation (UVPD) spectra (which provide radical *w*- and *d*-ions). By conditioning the flow on both dissociation modalities simultaneously, DFlowNovo can achieve 100% isobaric disambiguation without algorithmic guessing.

6.3.2 Orthogonal Analytical Dimensions (RT and CCS)

Modern mass spectrometers increasingly incorporate liquid chromatography retention time (RT) and trapped ion mobility spectrometry (TIMS) collision cross section (CCS). Because retention time correlates with peptide hydrophobicity and CCS reflects the gas-phase three-dimensional conformation of the ion, incorporating RT and CCS predictors as auxiliary energy terms during flow matching will further constrain the candidate sequence space.

6.3.3 End-to-End De Novo Antibody and Immunopeptidome Assembly

Finally, integrating DFlowNovo into an end-to-end overlap-graph assembler will enable the direct *de novo* sequencing of full-length monoclonal antibodies, circulating immunoglobulin repertoires, and tumor-specific neoantigens directly from patient biofluids without requiring patient genomic sequencing, accelerating the development of personalized immunotherapies.

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Finally, I dedicate this work to my family, whose love, patience, sacrifices, and unconditional belief in my aspirations have been my constant source of strength and motivation.

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